

POST-EVENT
DRAFT



BEng (Hons) Artificial Intelligence and Data Engineering (IIT)
School of Computing, Engineering and Technology

Version	1.0
Course Type	Undergraduate
Award Type	Bachelor of Engineering with Honours - BEng (Hons)
Awards	No Course Award Assigned
Code	0757
SCQF Level	Level 10
QAA Subject Benchmark	Computing (2022)
SCQF Credit Points	480
ECTS Credit Points	240
Valid From	2026 September

HECoS
The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
100160	computer aided engineering	computer aided engineering	50%
100359	artificial intelligence	artificial intelligence	50%
			Total 100%

Professional, Statutory and Regulatory Body (PSRB)
No PSRBs have been recorded

	The course will normally involve a four-year period which includes one-year industrial placement. Table 1 below shows how the course addresses a variety of subject streams to provide a wide range of knowledge and skills. <i>Table 1: Subject Streams of the Syllabus</i> <table><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>							

	Stream	Year	Semester	Subject	Core/ Optional	Credits
	Systems	1	1	Computer Systems Fundamentals	Core	15
				Database Systems	Core	15
			2	Web Technology	Elective	15
		3	2	Internet of Things and Wearables	Optional	15
				Edge Artificial Intelligence	Elective	15
	Programming	1	1 & 2	Programming Fundamentals	Core	30
			2	Data Structures and Algorithms for Artificial Intelligence	Core	15
		2	1	Object Oriented Development	Elective	15
	Mathematics	1	1 & 2	Computational Mathematics	Core	30
		2	1	Advanced Mathematics	Core	15
			1	Machine Learning	Core	15
				Artificial Intelligence	Core	15
				Data Engineering	Core	15

What the Course Involves	Data Engineering and AI	2	2	Simulation and Modelling Techniques	Elective	15
				AI Systems Deployment and Integration	Elective	15
				Agentic Artificial Intelligence	Elective	15
		3	1 & 2	Data Engineering Group Project	Core	30
			1	Deep Learning	Core	15
			2	Artificial Intelligence for Robotics	Elective	15
		4	1 & 2	Individual Research Project	Core	60
				Research Trends	Core	30
			1	Computational Intelligence	Core	15
				Artificial Intelligence for Entrepreneurship	Elective	15
				Machine Vision	Elective	15
				Natural Language Processing	Elective	15
	Training	3	1 & 2	Industrial Placement	Core	90
	Language	1	1	English Communication Skills	Core	[15]

In each year there will be two semesters. There will be two cohorts (in September and January) enrolling each year to the first year of the course. To accommodate both cohorts, delivery will be in September, January and May, i.e. the September intake will complete their second semester in April and the January intake will complete their second semester in August of the same year. However, the two cohorts commence their second year together in September, progress together through the other years and complete the course at the same time. This means the student joining in both September and January intakes of the same academic year could graduate together as long as they successfully complete all required credits. The curriculum of the course is structured so that prerequisites and core/optional modules are covered in a logical manner and gradually developing the knowledge and skill of the student.

	Students will attend eight semesters spread over four years. Each module has its own assessment pattern. Students will attend a minimum three hour (maximum four hour) session each week per module. A week will normally have four modules being delivered on campus and online. Lectures will be scheduled on weekdays from 8:00 am to 5:30 pm (Sri Lankan time) as and when required classes will be held on weekends. During the 3rd year while on industrial placement; lectures, tutorials and workshops will be delivered online and on campus. The course is made up of modules worth 15 credits, 30 credits, 60 credits and 90 credits (industrial placement). In the first and second year students would have a combination of 15 and 30 credit modules. However, in the third year, the industrial placement counts for 90 credits and in addition, students take two 15 credit modules. In the final year, students take four modules including a substantial individual research project with 60 credits. Every year students will be required to take a total of 120 credits for the course. A 15 credit module requires 150 hours of work – usually 50 hours will be scheduled activities at the IIT for the modules delivered on campus and during the online sessions for the modules taught online; lectures, tutorials, online activities and personal meetings, with the remaining 100 hours being private independent study. A 30 credit module requires 300 hours of work – usually 100 hours will be scheduled activities at the IIT; lectures, tutorials, online activities and personal meetings, with the remaining 200 hours being private independent study.
UCAS Code	Code
Course Collaborations	
Informatics Institute of Technology, Sri Lanka (Validated Course)	
Introduction to the Course	The course is a collaboration between Robert Gordon University (RGU) in Scotland and the Informatics Institute of Technology (IIT) in Sri Lanka. IIT design, deliver and assess the course and RGU quality assure and confer awards for the course. There is a rise in the demand for a workforce with skills in artificial intelligence and data engineering with the rapid enhancement of computer technologies in the 21st century. However, there is a skills shortage to meet the industry demands in this domain. This course is designed to address this shortage, as well as provide the opportunity for students aspiring to a career in artificial intelligence and data engineering. Students will be able to gain practical understanding of contemporary data engineering trends and technologies in developing cutting-edge data engineering solutions. Specifically, the course adopts a general to specific organisation of modules; whereby initially a range of modules organised under key streams; systems, programming and artificial intelligence and data engineering is gradually specialised into modules as students progress through Stages 1 to 4. More broadly, all modules can also be categorised into key knowledge categories such as programming, algorithms, machine learning, mathematics and statistics, where students are exposed to both theoretical principles and cutting-edge technological advances. This will enable students to unlock the power of automation, analytics and artificial intelligence, paving the way for innovation while on the course and after graduation. Industrial placement during the third year is a unique and mandatory feature of this course. Providing students the opportunity to gain industrial experience in artificial intelligence and data engineering through hands-on engagement with trending data science/engineering technologies, their applications and best practices to solve real world problems. Additionally, in the final year students are provided with the opportunity to specialise and conduct research in significant areas associated with artificial intelligence and data engineering as per the students' preferences via applying their knowledge gained through the taught aspects of the course and the practical exposures gathered during the placement year. The course allows the student to plan their academic learning effectively. In addition to on campus engagement for lectures, tutorials and lab sessions, with RGU's supported online learning platform and tools, students have flexibility to engage in their academic studies around personal commitments. Some of the modules taught in the course which includes CM1601 Design Lectures, CM1607 English Communication Skills, CE2603 Data Engineering Group Project, and the modules delivered in Stage 3 for placement students (CM3604

	Deep Learning, CM3602 Internet of Things Wearables and CM3603 Edge AI, CE3607 Artificial Intelligence for Robotics) will delivered online. Online learning can be accessed 24/7, relevant to the modules undertaken and includes the use of multimedia interactive activities, podcasts, quizzes, webinars and forum discussions, thus providing an engaging and motivating learning experience. Students are supported through the course to achieve professional excellence through a suite of modules designed to develop academic, personal and interpersonal skills. They will be encouraged to develop an enquiring, analytical, open minded and creative approach to foster independent judgement, critical self-awareness, the ability to reflect, and an appreciation of the wider context of the course of study. These skills will be developed and practiced both in the context of the academic environment and in their placement workplace. Ultimately, by generating a deep interest and enthusiasm for artificial intelligence and data engineering, students and graduates will be motivated to continually update their knowledge and understanding within the context of a rapidly changing technological, political, social and economic environment. Depending on the nature of the organisation, typical examples of career opportunities across a broad range of sectors, might include: data/enterprise architect, data engineer, artificial intelligent engineer, application architect, data scientists, machine learning scientist, machine learning engineer, data analyst, cloud/platform engineer (data & artificial intelligent systems), machine learning operational engineer, edge artificial intelligent / IoT Engineer, artificial intelligent systems, developer, software engineer (artificial intelligent focused) and an array of specialist positions. However, in many organisations, there may be specific elements of the course content involved in the role which are not reflected in the job title but which are a key component within it.
	The course aims to: • Design, develop, and evaluate robust computational and engineering solutions to address complex real-world problems using artificial intelligence and data engineering principles. • Apply mathematical, statistical, and computational techniques to analyse, model, process, and derive insights from large-scale and complex datasets. • Demonstrate competence in programming, software engineering, database systems, data structures and algorithms, artificial intelligence, machine learning, data engineering, and AI system deployment to develop intelligent and scalable solutions. • Evaluate, visualise, communicate, and interpret analytical and engineering outcomes using appropriate scientific, technical, and professional methods. • Formulate, design, implement, deploy, and manage medium- to large-scale artificial intelligence and data engineering projects using industry-standard tools, frameworks, and practices. • Apply ethical, professional, and sustainable engineering principles in the design, development, deployment, and governance of AI-driven systems. <u>Aims of Each Stage of the Course</u> Stage 1 (Certificate of Higher Education) • To demonstrate knowledge and understanding of fundamental concepts in programming, database systems, computer systems fundamentals, web technologies, computational mathematics, statistical techniques for data analysis, and data structures and algorithms that underpin artificial intelligence and data engineering. • To develop practical problem-solving, analytical, and computational thinking skills through theoretical and hands-on activities. • To establish a strong foundation in software development and computing systems required for advanced study in artificial intelligence and data engineering. Stage 2 (Diploma of Higher Education) • To demonstrate understanding of the fundamental concepts, techniques, and applications of artificial intelligence, machine learning, and data engineering. • To develop software development, object-oriented design, systems modelling, AI integration, and deployment skills required for building intelligent real-world applications.

Educational Aims of the Course	• To apply mathematical, computational, and engineering principles in the design, implementation, and evaluation of AI-enabled systems. • To design, develop, and evaluate a team-based data engineering solution by applying appropriate data engineering, artificial intelligence, and software development methodologies, tools, standards, and collaborative working practices. Stage 3 (BEng) • To demonstrate an in-depth knowledge and understanding of how artificial intelligence, data engineering, edge AI, IoT systems, and intelligent technologies are applied within industrial and research environments. • To critically evaluate problem-solving approaches, system architectures, and engineering solutions in real-world contexts. • To develop professional, ethical, and practical skills through industrial training and collaborative activities. • To demonstrate an understanding of the societal, ethical, economic, and technological impacts of artificial intelligence and data-driven systems. • To explore advanced areas of AI through specialised topics such as deep learning, edge AI, IoT, and AI robotics. Stage 4 (BEng (Hons)) • To demonstrate adaptability, innovation, and lifelong learning skills necessary for employment and leadership within the rapidly evolving fields of artificial intelligence and data engineering. • To undertake independent research and development through the design, implementation, and evaluation of a substantial individual project informed by current research and industry practice. • To develop advanced AI-driven and data-intensive systems by applying contemporary techniques, frameworks, and engineering methodologies. • To critically evaluate the feasibility, performance, security, scalability, reliability, and usability of intelligent systems deployed across a range of application domains and platforms. • To evaluate and utilise appropriate tools, frameworks, and engineering practices for the planning, design, implementation, deployment, monitoring, and maintenance of AI and data engineering solutions. • To demonstrate effective teamwork, project management, leadership, entrepreneurship, and professional communication skills while contributing to complex engineering projects and innovation activities.
Awarding Institution	Robert Gordon University
Course Duration	Full time: 4 years. The course has two intakes in September and January. The maximum period of enrolment for the course is 6 years.
	BEng (Hons) Artificial Intelligence and Data Engineering is a four-year course, with a one-year full-time placement in its third year. A credit bearing placement year is a unique selling point as students acquire one year of work experience prior to graduation. This enriches their final year experience and at the same time increases the chance of securing employment on graduation. The course content provided in a logical manner gradually develops students' knowledge and skills in artificial intelligence and data engineering. This will enable them to become experts in two complementary disciplines which have a high demand locally and globally.

Distinctive Features of the Course	Main unique features include: • A substantial project module in second year, in the form of a data engineering group project. This will allow the student to practically experience the theoretical aspects learnt in a group environment collaboratively. It will give students a competitive edge enabling them to secure placement opportunities in the industry in third year, while giving them an exposure in artificial intelligence and data engineering applications development and its lifecycle prior to their industrial placement. This will allow the student to practically experience theoretical aspects learnt in a group environment collaboratively. It will also act as an early-stage eye-opener for the student to develop their research-oriented insight in a group setting resolving inter-group conflicts as well. • As mentioned, the credit bearing industrial placement in Stage 3 is a unique feature of this course. It aims to align with the IIT's unique selling point of delivering courses that are relevant to industry. The course will benefit from the IIT's strong collaborative links with industry partners. This creates an effective practice-led learning environment for students and also complements the highly skilled panel of lecturers at the IIT. The placement year provides the opportunity for the students to practically apply the concepts they have learnt in the first two years of the course. Additionally, the placement atmosphere exposes the students to real life working culture. A student who completes the placement year will improve their skills in problem-solving, customer handling, time management, peer liaising and work-life balance, which are valuable attributes for graduates. • Methodical organisation of the syllabus, where multiple streams (programming, systems and artificial intelligence and data engineering) is progressively developed by introducing increasing levels of complexity to the list of modules organised and gradually moving towards a specialisation in the artificial intelligence and data engineering stream. This will ensure versatile and enriched data engineering-related content coverage. • Case studies of current industry trends as well as research in state universities will be showcased through invited talks feeding into research trends and other Stage 4 modules. Content covered in the course provides a strong foundation on theoretical and practical skills, and at the same time nurtures creativity and innovation through teaching and learning methods. This facilitates students to compete at local and international competitions, forums and conferences developing confidence and ability for future careers. • The programme incorporates specialised elective pathways that enable students to pursue areas of particular interest within artificial intelligence and data engineering. This flexibility allows students to develop expertise in emerging and industry-relevant domains while maintaining a strong foundation in core principles. As a result, graduates are well prepared for a diverse range of careers and are able to adapt to rapidly evolving technologies and professional practices. • The profile of the lecturers involved in delivering this programme is unique in that it includes a range of academics as well as experts drawn from industry. All of them are PhD, MPhil or Masters qualification holders with relevant industry and research experiences, blended with required teaching experience. The course uses a multidisciplinary approach to delivery, without being limited to traditional lectures on campus. Approaches like workshops, guest lectures, industry visits, webinars, and hackathons will also be used aided with new cutting-edge teaching methods (i.e. e-learning techniques, online teaching and learning experience, hands-on practical flipped classrooms) to improve overall efficacy. Thus, students become familiar with new communication and online learning methodologies, which is good for application in industry, as well as an enhancement to their learning experience, since the industry also promotes self-learning via new online learning techniques.
Placement/Professional Experience	Students enrolled on the course are required to undertake an industrial placement during the third year of the course. The IIT works closely with industry bodies such as Federation of Information Technology Industry Sri Lanka (FITIS) and Sri Lanka Association of Software and Services Companies (SLASSCOM), two associations covering all major IT companies in Sri Lanka, and many of its alumni are linked to leading software companies in the country (Virtusa, WSO2, MilleniumIT, 99X Technologies, IFS). These strong links with industry as well as the increased need for artificial intelligence and data engineering application developers locally and globally provides excellent opportunities for industry placements and Honours projects that are driven by applied problems. Additionally, guest lectures, workshops led by industry specialists, will be another opportunity for the undergraduate students to update themselves as well as to establish new networking opportunities. Since students who have completed the placement should have developed strong links with the industry specialists in specific areas, they can use this network to invite experts to the University to conduct workshops and knowledge sharing sessions on specialised technologies/work procedures.

	The IIT has clubs to represent professional bodies such as IEEE, IET and Rotaracton campus. IEEE has three clubs namely IEEEExtreme, Computer Science and Women in Engineering. The on campus clubs facilitate students to work with industry experts, undergraduates from other universities, participate in technical forums, conferences and workshops to enhance their knowledge and skill and develop a network with professionals locally and globally.
Opportunities for Further Study	An Honours degree achieved with at least an upper second classification (2.1) will normally provide access to postgraduate and doctoral programmes in UK higher education institutions. In addition, in Sri Lanka and other countries (e.g. USA, Australia), some of the top ranking universities require a four-year degree with a credited placement as paramount to enter postgraduate degrees. Since the BEng (Hons) Artificial Intelligence and Data Engineering meets these requirements, students have the opportunity to pursue such postgraduate degrees. Furthermore, the course has a strong grounding in research techniques and enables students to advance into research and development at postgraduate and doctoral levels. RGU and the IIT are exploring this avenue of jointly supervised PhD opportunities with matched funding schemes. Increasingly, doctoral programmes by online learning are growing in number, format, subject area and modes of delivery.
External and Internal Indicators of Quality and Standards	QAA Subject Benchmark Statements are part of the QAA Quality Code. They set out the nature of study and the expectations of graduates in specific subject areas. All undergraduate and postgraduate courses delivered at the IIT in collaboration with its partner institutions are in alignment with a Subject Benchmark Statement in their subject, and for this course it is Computing. Scottish Credit and Qualification Framework (SCQF) statements are also used during the design of modules, courses and programmes. The relevant SCQF level is included within each Module Descriptor. RGU appoints External Examiners to assure standards of assessment. Examiners are appointed from both academic and practice backgrounds to cover key subject areas. The External Examiner will verify that standards are appropriate for the award elements, assist in the comparison of academic standards across higher education awards and ensure that assessment processes are fair and equitable in line with RGU's Regulations. Validation and annual monitoring are carried out by partner institutions to ensure that the IIT meets the standard. British Computer Society (BCS) accreditation of courses provides an indicator of quality to students and employers and provides independent recognition for institutions that offer them. BCS has accredited all undergraduate and postgraduate courses delivered by the IIT including two MSc programmes delivered in collaboration with RGU. The IIT obtained certification in 2019 to ISO 9001: 2015 Quality Management Systems (QMS), which is internationally recognised as the world's leading quality management standard and has been implemented by over one million organisations in over 170 countries globally. Annual audits are carried out to ensure compliance. The IIT works closely with industry bodies such as Federation of Information Technology Industry Sri Lanka (FITIS) and Sri Lanka Association of Software and Services Companies (SLASSCOM), two associations covering all major IT companies in Sri Lanka, to ensure that the curriculum is up-to-date and meets the employee expectations. In 2018 the IIT took part in the SLASSCOM hot skill audit programme and was ranked within the top 10 universities in Sri Lanka.
	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular:

University Academic Regulations	Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards								
University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.								
	IIT will adopt RGU's Academic Regulations with the following agreed adaptations and amendments: <u>Regulation A2: Admission</u> Admission Requirements for Undergraduate Courses Firstly, all students are required to obtain one of the following: <table border="1" data-bbox="450 804 2087 986"> <tr> <th data-bbox="450 804 1252 892"><u>Sri Lankan Requirements</u></th><th data-bbox="1252 804 2087 892"><u>UK Requirements</u></th></tr> <tr> <td data-bbox="450 892 1252 986"> - Five GCE/O Level passes with a minimum grade C in English and Maths </td><td data-bbox="1252 892 2087 986"> - Six GCSE passes with a minimum grade C in English and Maths </td></tr> </table> The Sri Lankan requirement of an O Level pass in English accounts for the English language requirements for the course. To build on this, CM1607 <i>English Communication Skills</i> is a core module delivered in Semester 1 of Stage 1 to facilitate students in basic written/verbal communication skills needed for the course. Successful completion of the module is required for progression to Stage 2 but the 15 SCQF credits do not contribute to any award. Secondly, all students are required to obtain one of the following: Standard Admission Requirements <table border="1" data-bbox="450 1305 2087 1487"> <tr> <th data-bbox="450 1305 1252 1393"><u>Sri Lankan Requirements</u></th><th data-bbox="1252 1305 2087 1393"><u>UK Requirements</u></th></tr> <tr> <td data-bbox="450 1393 1252 1487"> - Three A Level passes with a minimum grade S in a science stream </td><td data-bbox="1252 1393 2087 1487"> - Three A Level passes with a minimum grade D in a science stream </td></tr> </table>	<u>Sri Lankan Requirements</u>	<u>UK Requirements</u>	Five GCE/O Level passes with a minimum grade C in English and Maths	Six GCSE passes with a minimum grade C in English and Maths	<u>Sri Lankan Requirements</u>	<u>UK Requirements</u>	Three A Level passes with a minimum grade S in a science stream	Three A Level passes with a minimum grade D in a science stream
<u>Sri Lankan Requirements</u>	<u>UK Requirements</u>								
Five GCE/O Level passes with a minimum grade C in English and Maths	Six GCSE passes with a minimum grade C in English and Maths								
<u>Sri Lankan Requirements</u>	<u>UK Requirements</u>								
Three A Level passes with a minimum grade S in a science stream	Three A Level passes with a minimum grade D in a science stream								

Course-Specific Entry Requirements	Non-Standard Admission Requirements			
	<u>Sri Lankan Requirements</u>			<u>UK Requirements</u>
	Two A Level passes with a minimum grade S in a science stream and a minimum of 12 points based on the below points scheme			Two A Level passes with a minimum grade D in a science stream and a minimum of 12 points based on the below points scheme
	Successful completion of the IIT Foundation Certificate in Higher Education Programme with at least 50% in the Mathematics for Computing and Business and Introduction to Programming modules			
	Partial completion of an Honours degree course in computing, conducted by any other recognised university, local or foreign			
	Points Scheme			
	A Levels			
	Sri Lankan		UK	
	Grade	Points	Grade	Points
	A	10	A	10
	B	8	B	8
	C	6	C	6
	S	4	D	4
			E	2

	As Sri Lankan A Level assessments are untaken in August, provisional Sri Lankan A Level grades can be used as a basis for an offer of admission providing the provisional grades meet the above specified admission requirements. When the formal Sri Lankan A Level results are released in December, students will be subject to the following conditions: 1. A student who does acquire the required A Level results as specified above continues on the course to Semester 2 in January. 2. A student who does not acquire the required A Level results, but has passed all of Semester 1 modules, should leave the course but their Semester 1 performance is retained. The student could return to Semester 2 of the course in the following year after successful completion of the IIT Full Foundation Programme from January to August. 3. A student who does not acquire the required A Level results and has failed all or some of Semester 1 modules, should leave the course but their Semester 1 performance is retained. The student could return to Semester 2 of the course in the following year after successful completion of the IIT Full Foundation Programme from January to August. The student would be required to undertake any reassessments from Semester 1 in August. On successful completion of Semester 2 and any reassessments, the student can continue on the course to Stage 2 in September with their original cohort.
Alternative Qualifications for Entry	Applicants with qualifications in other streams, such as the technology stream, will be considered on a case-by-case basis by the Course Team.
Interview Procedures	Refer Course Specific Entry Requirements.
Course-Specific Regulations	<u>Regulation A3-1 Academic Appeals (Awards and Progression) Procedure</u> Regulation A3 - Section 1 shall apply to all enrolled students on the course. <u>Regulation A3-2: Student Misconduct Procedure</u> The Student Misconduct Procedure is replaced by the IIT's 15 Misconduct or Malpractice Procedure – Students. <u>Regulation A4: Assessment and Recommendations of Assessment Boards</u> Approval of Instruments of Assessment Assessments will be internally moderated by the IIT, overseen by RGU as appropriate and sent to the External Examiner for approval. Assessment Board The Assessment Board membership shall be constituted in accordance with Regulation O7: Assessment Boards and shall be approved by the RGU School Academic Board. The membership shall normally comprise a Convener (RGU Dean), Dean (IIT), External Examiner, RGU Link Coordinator, IIT Head of Department, IIT Course Leader, IIT Module Coordinators (on request) and a member of staff from another School (RGU). Representatives from IIT Registry and RGU Academic Quality shall be in attendance.

	The Assessment Board will be convened twice each year to decide on student progression and award. Condonement CM3606 <i>Industrial Placement</i> and CM4609 <i>Individual Research Project</i> are central to the objectives of the course and, therefore, cannot be condoned. <u>Complaints Procedure</u> Complaints are managed through the IIT QMS 16A Complaint Handling Procedure. CM1607 English Communication Skills is a core module, which is required for progression to Stage 2 but the 15 SCQF credits do not contribute to any award.
Equality, Diversity and Inclusion	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU. All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.
University Strategy	The Purpose of the University is: - To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is: - To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability. The Culture of the University is: - To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality. Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background. At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK.

	RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region. RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.
Course-Specific Learning Experience	Course Philosophy and Approach to Learning The programme adopts a student-centred and practice-oriented approach that combines theoretical foundations with authentic learning experiences. Students are encouraged to develop technical expertise, critical thinking, problem-solving abilities, and professional skills through a combination of lectures, tutorials, laboratory activities, projects, and independent learning. The curriculum emphasises real-world applications, collaboration, research-informed teaching, and the development of lifelong learning skills relevant to the rapidly evolving field of Artificial Intelligence and Data Engineering. Learning Environment Students benefit from a supportive learning environment equipped with modern lecture theatres, tutorial spaces, computer laboratories, and teaching technologies including projectors and digital learning platforms. Where required, online teaching and support are facilitated through platforms such as Zoom. Students have access to a range of digital resources including the Virtual Learning Environment, library services, electronic databases, and LinkedIn Learning to support both academic and professional development. Individual and Group Learning The programme incorporates both individual and collaborative learning experiences. Individual learning is promoted through assessments such as the Individual Research Project and independent coursework activities, while teamwork and collaborative skills are developed through modules such as the Data Science Group Project. Students also participate in presentations and communication-focused activities, including those embedded within English Language modules, enabling them to develop professional and interpersonal skills. During the industrial placement year, students continue to engage in structured learning through Institute-delivered modules such as Professional Development, which support reflective practice, employability, and professional growth. Study Visits and Trips Students may participate in field visits and external engagements where appropriate to support learning outcomes. Modules such as the Data Science Group Project may involve visits to organisations or stakeholders, subject to institutional approval and guidance. Students undertaking research projects may also conduct fieldwork and data collection activities, with the Institute providing the necessary approvals and support to facilitate these activities safely and effectively. During the industrial placement period, academic staff conduct placement visits and maintain regular engagement with both students and employers to monitor progress and provide academic and pastoral support. IT Provision and its Place in the Learning Process

Technology forms an integral part of the learning experience. Students have access to computing facilities, specialist software, digital libraries, and online learning resources that support both teaching and independent study. Additional support mechanisms include placement preparation and mentoring, academic supervision, and guidance for external competitions and innovation activities such as NBQSA. Throughout the industrial placement year, students receive continuous support through placement visits, regular communication with academic mentors, and structured professional development activities designed to enhance employability and workplace readiness.

Developing Entrepreneurial and Innovative Practice

At Stage 1, the English Communication Skills module supports the development of communication, presentation and professional skills that provide a foundation for students to articulate and present ideas effectively. At Stage 2, entrepreneurial and innovative practice is deliberately embedded within the Data Engineering Group Project, where students work collaboratively to identify real-world problems and develop viable data/AI-driven products or solutions. Through project development, demonstrations and viva assessment, students develop teamwork, problem-solving, innovation and the ability to communicate and justify their proposed solutions. At later stages of the course, this is further developed through the industrial placement, providing exposure to real-world professional practice, and the Individual Research Project (IRP), where students independently investigate contemporary problems and design, implement and evaluate innovative solutions. Beyond the formal curriculum, students are encouraged to showcase their work through IIT's annual Cutting Edge exhibition, while IIT LEAP Gen, an entrepreneurship and acceleration initiative, provides opportunities for students interested in progressing innovative ideas towards entrepreneurial ventures. The course further strengthens this progression through the introduction of Artificial Intelligence for Entrepreneurship as an elective in Stage 4, providing a dedicated pathway for students wishing to develop their entrepreneurial capabilities further. Overall, the course provides a progressive journey from communication and idea development, through collaborative product development and professional practice, to independent innovation and entrepreneurship.

Teaching Strategies and Self-Learning

BEng (Hons) Artificial Intelligence and Data Engineering is a novel course, embracing both artificial intelligence and data engineering pathways in catering to the prevailing skills shortage. A variety of trending and effective teaching and learning strategies are utilised to deliver the contents of the course to the enrolled students. These strategies are inclusive of formal face-to-face lecture series, several online lectures, guided laboratory sessions, tutorial sessions on campus and online, workshops, guest sessions by industry experts, field visits as well as student-centred exhibitions, competitions and hackathons.

Both lecturer-centred and student-centred approaches are used for enforcing bi-lateral interaction between both the groups. At the inception for the first-year students, more attention and follow up is required, as they are in a transition period of getting adjusted to the university education system. Therefore, at first year, more lecture-centred emphasis is provided and gradually, when the student moves from second to third and fourth year, student-centred emphasis can be increased. Through these mechanisms, students will learn about self-responsibility and how to efficiently manage the autonomy given for them. These skill-sets are exclusively important when the student enters the society as a matured professional.

In addition to the guided learning strategies, students are encouraged to develop their individual learning as well as industry always gives prominence to quickly adaptable, research-oriented, self-learning individuals who can work with minimal or no supervision.

Skills Development

The employability skills students require are not limited to discipline-specific hard skills as soft skills also play a vital role in uplifting an individual to the state of a professional. Each subject stream contributes to the enhancement of the students' skillsets. Those skills could be numerical skills, which are effectively addressed from the mathematics stream associated modules. Soft skills such as communication, conflict resolution, group interactions, time management, autonomy accountability and cognitive aspects are addressed with multiple assessments comprising group work activities, presentations, thesis/report writing. Additionally, strong emphasis is given to improve the technical skillsets required to perform as a skillful Software Engineer or a specialised Data Scientist.

Teaching and Learning Strategy	Learning Environment Upon enrolling to the course, a student will obtain access to the RGU's Moodle, e-learning environment. This is used to place course content, assignments, and liaise between students and Module Lecturers. Lecturers can upload course material, create discussion forums, quizzes and numerous forms of learning and testing experiences are available to the students via RGU's e-learning environment. In addition, after enrolling in the course, each student will be given an RGU student account. Through the credentials of this account, students will be able to access almost all established research repositories (i.e. IEEE Xplore, Springer, ACM) which will be an ideal platform to enforce a researched-oriented culture among students. At the IIT, students are equipped with state-of-art lecture halls, lab facilities with modern software and an up-to-date library which contains most of the books/articles related to the programme delivered. Both labs and library are accompanied by supportive staff to facilitate the student requests. Further, the course is taught by experienced staff with necessary technical, teaching and research expertise. Another strength of the IIT is the alumni as the IIT has more than 3,000 alumni, who are in distinguished roles and who could contribute to the course by sharing their expertise, strong industry links, supervising research projects and various other means. Student-Lecturer Interaction Constant and appropriate student–lecturer interaction is vital in transforming a student to accomplish the graduate attributes. In addition, to formal email communication alternative communication methods depending on their needs, such as face-to-face, email, chat, virtual learning environment forums, Skype and phone are also facilitated. The list below describes how various hierarchies related with the course communicate with students. • A Module Leader: module related queries (via Email, Moodle, face-to-face at Lectures/Tutorials/Labs) • The Course Leader: course queries and/or personal issues (via Email, Moodle, Phone, face-to-face) • Student Relations Manager: non-academic matters and personal issues (via face-to-face) • The Registrar: module registration related issues. (via Email) • IT Services: for technical support in the laboratories and/or for setting up software on their own devices e.g. Wi-Fi (via Email) • Library staff (via Email) • Designated email contact at RGU: course validity related questions (via Email)
	A variety of assessment methods are used in this course, with particular emphasis placed on coursework assignments, including design exercises (ranging from design studies to fully implemented solutions), comparative studies, investigative reports, demonstrations, presentations, interviews, computer aided assessments and unseen written examinations. Students are informed in advance what assessment methods will be used for each module. The format of assessment varies significantly from one year to the next, examples of assessment will be created in order to ensure students understand what the assessment entails. All assessments include a marking/grading scheme, which describes the criteria used for the award of marks/grades. Assessment Plan Coursework may be used as a form of formative and/or summative assessment, and within the context of industrial training this may be work-related. Coursework deliverables cover a variety of artefacts such as technical reports, presentations, demonstrations, assessed labs. These will be used to assess problem identification, problem analysis with evaluation of alternative solution strategies, solution design, evaluation of solution performance, documentation and presentation skills. With assessed labs, where students undertake a coursework in a computer laboratory under examination conditions, students will normally also have access to a range of reference materials. The use of state-of-

the-art computational tool kits and programming skills will be assessed in all practical coursework assignments in addition to analytical and critiquing skills (i.e. literature survey, evaluation methodologies) and communication skills (results presentation and documentation). Technical reports will be used to document coursework assignments. Demonstration of IT competence is mandatory in production of documentation to support all coursework submissions.

Within the industrial placement module, work-related coursework will be a portfolio of evidence showing engagement in the industrial placement, such as attendance in pre-placement training sessions, creation of a resume, personal and professional development plans, placement agreement, final placement report along with records issued by the company, such as the letter of service for the period of the placement, poster presentation and viva. Coursework could also be work-centric, such as a report in response to a case study scenario where the case study is built to reflect current issues within a specific sector.

With the individual research project, a plagiarism analysis report, such as from Turnitin will accompany the final dissertation. Supplementary documentation on project management will record project progress, quality guidelines and project deliverables. It will also contain information on the use of WWW resources to present interim individual research project progress updates and guidelines on the live presentation and demonstration of individual research project outcomes.

The following sections detail the assessment of the SCQF Level Descriptors.

Knowledge and Understanding

For taught modules the focus will be to assess the student's ability to integrate key features, boundaries and conventions through a critical understanding of the principal theories, concepts and principles. Students will be expected to have a detailed knowledge and understanding when addressing assessment components in each of the core technological modules. The methods of assessment used are mentioned above, such as coursework.

Practice: Applied Knowledge and Understanding

Practical skills are assessed throughout coursework assignments and examinations which involve the use of a variety of computing tools and packages as well as programming environments. Evaluation skills are key to demonstrating knowledge and understanding of business analytics practice. The individual research project will assess practical skills required to complete a substantial real-world research problem.

The technical modules will assess a range of specialised tools and packages that are at the forefront of computer science with a special focus of artificial intelligence and data engineering related development. Students will be expected to integrate knowledge and skills from a range of such offerings and techniques of inquiry to create appropriate solutions for assessment tasks.

Some of these modules will assess a range of professional skills, techniques and best practices as informed by legal, ethical and social aspects.

The opportunity to assess group working is also integrated within several modules comprising assessment components that have group-based coursework, in addition to the group project.

The integrative individual research project module is primarily designed to assess the student's ability to plan and execute a significant project of research, investigation and software development to demonstrate originality, creativity, and aptitude for innovation. Presentations and a viva component will further assess the student's ability to respond positively to critique and participate in peer-review processes. Importantly, it will also expose students to the unknown, and often unpredictable, variety that is to be expected of professional level contexts.

Generic Cognitive Skills

Assessment generally involves the production of technical report content, which reflects on problem identification, critical appraisal and analysis of alternative solutions.

With the technical modules, assessments will focus on testing the student's ability to critically analyse and evaluate issues in real-world scenarios.

The assessment of generic cognitive skills is particularly evident in the individual research project module. In particular, students will be expected to critically review, consolidate and extend knowledge, skills, practices and thinking in relation to their chosen research project. These projects provide the student with the opportunity to excel in their original and creative problem-solving skills. Although not a requirement, it has the potential to generate work that can lead to publishable research material.

Communication, ICT and Numeracy Skills

Communication skills are assessed through technical reports documenting development of assignments; structured project documents to manage project conduct, establish quality guidelines, archive project deliverables; use of web resources to present individual research project progress; live presentation of research project outcomes. Communication tasks are implicitly assessed in tasks which involve both oral/written communication as well as documentation.

ICT skills are assessed through the use of IT tools in all coursework assignments (i.e. use of tools to plan, aid design, collate results and document coursework). Demonstration of IT competence is mandatory in production of documentation to support all coursework submissions including development of support material for the project oral presentation. Numeracy skills are essential for the successful completion of most course modules and are implicitly assessed in coursework and unseen written examinations. Numeracy skills are particularly key to data analysis tasks and the evaluation of project results.

Autonomy, Accountability and Working with Others

Explicit assessment of these skills is evident in the project conduct of the individual research project, which reflects the student's interaction with the supervisor and includes demonstration of own-initiative and ability to work independently.

Other coursework that involves presentations will equip students to handle peer pressure and respond appropriately to questions. Team-working skills will be assessed in the modules which include group work for the compilation of a critical report on legal, ethical, professional and social issues.

Table 2: Mapping SCQF Characteristics to Modules

SCQF Level	Module	Module Code	C1	C2	C3	C4	C5
7	Database Systems Design and Implementation	CM1603	x	x	x		
7	Computer Systems Fundamentals	CM1604	x	x			

Assessment

7	Computational Mathematics	CM1606	x	x	x	x	
7	Programming Fundamentals	CM1601	x	x			
7	English Communication Skills	CM1607	x				
7	Web Technology	CM1605	x	x	x	x	x
7	Data Structures and Algorithms for Artificial Intelligence	CM1602		x	x		
8	Object-Oriented Development	CM2601		x	x		
8	Artificial Intelligence	CM2602	x	x	x	x	
8	Advanced Mathematics	CE2607	x	x	x	x	
8	Simulation and Modelling Techniques	CM2605	x	x	x		
8	Machine Learning	CM2604		x	x		
8	Data Engineering	CM2606		x	x		
8	Data Engineering Group Project	CE2603	x	x	x	x	x
8	AI Systems Deployment and Integration	CE2609	x	x	x	x	x
8	Agentic Artificial Intelligence	CE2608	x	x	x	x	

	9	Industrial Placement	CM3606	x	x	x	x	x
	9	Deep Learning	CM3604		x	x		
	9	Edge Artificial Intelligence	CM3603	x	x	x		
	9	Internet of Things and Wearables	CM3602	x	x	x		
	9	Artificial Intelligence for Robotics	CE3607	x	x	x	x	
	10	Individual Research Project	CM4609	x	x	x	x	x
	10	Research Trends	CM4604		x	x	x	x
	10	Computational Intelligence	CM4607		x	x		
	10	Machine Vision	CM4610	x	x	x		
	10	Natural Language Processing	CM4608	x	x	x		
	10	Artificial Intelligence for Entrepreneurship	CE4611	x	x	x	x	x
The University provides a number of means of supporting teaching and learning: • student induction organised on a course or School basis; • a Team Around a Student (TAAS) system of pastoral support to students including online support www.rgu.ac.uk/policies [The TAAS School Student Support Model replaces the Personal Tutor model of student support from Session 2026-27]; • an extensive programme of student study skills delivered through Student Life and the Library; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies;								

University Support for Teaching, Learning and Assessment	- the University's active participation in Scotland's Tertiary Enhancement Programme (STEP) www.step.ac.uk ; - support for staff from the Department of Academic Quality and Learning Enhancement (AQLE), including credit-rated teaching, learning and assessment provision; - extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; - access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub; - Moodle, http://campusmoodle.rgu.ac.uk , the University's dedicated virtual learning environment; - a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; - the expanding provision of state-of-the-art, purpose-built facilities and buildings.
Course-Specific Support for Teaching, Learning and Assessment	Academic support is provided by the academic staff delivering the course. The course is led by a Course Leader, who is the person to approach to ask about any aspect of the course. Each module is led by the Module Leader, a member of academic staff who, with colleagues, guides a student's study in a variety of ways from formal lectures to private conversation. Each student will be assigned a Personal Tutor to support them. In order to acquaint themselves with their Personal Tutor, students will meet them on a regular basis.
University Monitoring of Quality and Standards	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including: <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires; <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process; <i>External Examiner Annual Reports</i> ; - Ongoing liaison with industrial/professional liaison groups. Formal Committees with responsibility for monitoring and evaluating quality and standards: - Staff/Student Liaison Committees, or equivalent; - Course/Programme Management Teams; - School Academic Boards; - Assessment Boards; - Quality Assurance and Enhancement Committee and associated Sub-Committees.
	This validated course is delivered and assessed by the IIT and approved and overseen by RGU. All academic collaborations are subject to approval, monitoring and review by RGU. The ultimate responsibility for the course monitoring and maintaining the standard of the course lives with RGU.

Course-Specific Monitoring of Quality and Standards	A Link Coordinator is appointed to the course by RGU to take responsibility for the general operation, coordinator and monitoring of the collaboration. The Link Coordinator provides the academic link between RGU and the IIT, ensuring adequate ongoing communication between the institutions and acts as an adviser to the Course Team. The Link Coordinator produces a Link Coordinator Annual Report as part of the Annual Appraisal process. The collaboration is formalised in a Contract of Collaboration. Monitoring and compliance with contractual obligations is led by Business and Economic Development at RGU. The IIT also have a Quality Manual and Quality Policies in place.
University Student Feedback Mechanisms	The University and RGU:Union work in partnership to create an environment which stimulates: • the participation of students, by empowering them to proactively provide views and opinions; • a supportive learning community where students and staff engage in meaningful dialogue; and • the engagement of students in the design of solutions and enhancements. This approach is underpinned by the <i>Student Partnership Agreement</i> which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: Student Representation & Partnership RGU . There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience. Mechanisms through which students are engaged and supported include: • participation in student evaluation questionnairesand where relevant the <i>National Student Survey (NSS)</i> ; • staff/student liaison arrangements at course/programme level; • support for Student Representatives; • student representation on Institution-Led Subject Review Panels and Validations; • student representation on key University committees.
Course-Specific Student Feedback Mechanisms	Feedback from the IIT will be given to students in response to assessment, in response to questions in lectures, seminars and tutorials, and in guidance received during supervision. Students receive feedback each semester through a number of mechanisms: • Formal feedback on summative assessment. • Regular in-class feedback such as verbal feedback in tutorials, peer feedback and feedback during Q&A sessions. • Feedback via project supervision meetings when students are undertaking project-based modules.

- Industry feedback during industrial visits when students are undertaking the Industrial Placement module.

Awards

Stage 1

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Certificate of Higher Education - CertHE	Artificial Intelligence and Data Engineering	Aberdeen, United Kingdom	Exit Award		120

Stage 2

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Diploma of Higher Education - DipHE	Artificial Intelligence and Data Engineering	Aberdeen, United Kingdom	Exit Award		240

Stage 3

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Bachelor of Engineering - BEng	Artificial Intelligence and Data Engineering	Aberdeen, United Kingdom	Exit Award		360

Stage 4

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Bachelor of Engineering with Honours - BEng (Hons)	Artificial Intelligence and Data Engineering	Aberdeen, United Kingdom	Exit Award		480

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

#	Description	SCQF Characteristic	SCQF Level
LLO1	Demonstrate a basic knowledge and understanding of mathematics, web technologies, fundamentals of computer systems, database systems design and implementation and basics of programming.	SC1	SCQF 7
LLO2	Demonstrate knowledge and understanding of mathematical principles required for solving data-driven and data engineering problems, forming the foundation for AI systems and data engineering applications.	SC1	SCQF 7
LLO3	Demonstrate the basic understanding of computer science principles and related subject disciplines. The student is introduced to the required skills and knowledge for employing basic tools and techniques.	SC1	SCQF 7
LLO4	Apply knowledge, skills and understanding of mathematical principles, web technologies, database design and implementation in practical contexts.	SC2	SCQF 7
LLO5	Study in a systematically directed way with the aid of appropriate tutor guidance within the classroom and laboratory. All topics include elements, which require and encourage the development of these skills.	SC2	SCQF 7
LLO6	Carry out routine inquiry, development or investigation into professional level problems and issues.	SC2	SCQF 7
LLO7	Apply basic computer science and mathematical principles following procedures along with basic tools and techniques to solve standard, simple problems.	SC3	SCQF 7
LLO8	Demonstrate the ability to analyse a data-driven problem and visualise data based on the specification provided.	SC3	SCQF 7
LLO9	Display the ability to form one's own assessment based on criteria provided.	SC3	SCQF 7
LLO10	Demonstrate written and oral communication skills required to describe computer science aspects in an understandable and concise manner.	SC4	SCQF 7
LLO11	Demonstrate practical skills in web technologies and database design and implementation using basic ICT applications.	SC4	SCQF 7
LLO12	Demonstrate numerical and graphical skills necessary for data processing and data manipulation in data engineering contexts.	SC4	SCQF 7
LLO13	Exercise some managerial or supervisory responsibility for the work of others within a defined and supervised structure.	SC5	SCQF 7
LLO14	Lead in implementing agreed plans in familiar or defined contexts, while taking account of your own and others' roles and responsibilities when carrying out and evaluating tasks.	SC5	SCQF 7
LLO15	Work, under guidance, with others to acquire an understanding of current professional practice, managing limited resources within defined areas of work.	SC5	SCQF 7
LLO16	Demonstrate the knowledge associated with advanced mathematical theorems for artificial intelligence and data engineering and define intelligent problem-solving methods and their applications associated with philosophical and cognitive theory in artificial intelligence.	SC1	SCQF 8
LLO17	Demonstrate knowledge and understanding of various modelling techniques, types of processes and probabilistic techniques in data modelling.	SC1	SCQF 8
LLO18	Demonstrate a discerning understanding in identifying and comparing contemporary software development methodologies, project management methodologies, quality assurance standards related to innovation for producing data engineering solutions.	SC1	SCQF 8
LLO19	Demonstrate an awareness and understanding of ethical, social, professional and legal issues associated with managing datasets and use of machine learning models in the real-world.	SC1	SCQF 8
LLO20	Apply knowledge, skills and understanding in identifying, contrasting and applying advanced theories, concepts, specific tools and techniques for solving real world problems in artificial intelligence and data engineering.	SC2	SCQF 8

LLO21	Apply knowledge, skills and understanding in selection of an appropriate evaluation technique and the application of appropriate data mining techniques to a given problem.	SC2	SCQF 8
LLO22	Apply knowledge, skills and understanding in using a range of professional skills, techniques, practices and appropriate tools in line with suitable software development methodology to meet a designated set of requirements and standards.	SC2	SCQF 8
LLO23	Compare and contrast reasoning and knowledge representation strategies used in artificial intelligence and data engineering systems.	SC3	SCQF 8
LLO24	Use a range of approaches to formulate and critically evaluate, contrast and apply suitable techniques related to artificial intelligence problem solving.	SC3	SCQF 8
LLO25	Apply appropriate techniques related to advanced mathematical concepts and theorems and apply them to simulate the real-world processes.	SC3	SCQF 8
LLO26	Analyse user centred design process, cognitive aspects, research methods, modelling and prototyping used to produce applications related to solving real-world data-driven problems.	SC3	SCQF 8
LLO27	Analyse legal, ethical, and professional issues in the context of intelligent systems solutions for real-world applications.	SC4	SCQF 8
LLO28	Apply a selection of advanced numerical, modelling and design skills, relevant to specialised areas of data mining, simulation, data engineering systems, data driven software solutions and artificial intelligence.	SC4	SCQF 8
LLO29	Use a wide range of specialised skills, techniques and standard applications associated with conveying complex information as well as evaluate numerical and graphical data for real-world applications related to artificial intelligence and data engineering.	SC4	SCQF 8
LLO30	Develop skills necessary to collaborate within a team on a data engineering and AI systems development project.	SC5	SCQF 8
LLO31	Exercise autonomy and initiative at a professional level in practice for the work of others within a defined structure.	SC5	SCQF 8
LLO32	Work under guidance with others to acquire an understanding of current professional practice and managing resources within defined areas of work.	SC5	SCQF 8
LLO33	Manage under guidance ethical and professional issues in accordance with current professional and/or ethical codes or practices.	SC5	SCQF 8
LLO34	Demonstrate a critical understanding of the strengths and limitations of artificial intelligence and deep learning technologies and autonomous systems including robotics, and an awareness of related ethical issues.	SC1	SCQF 9
LLO35	Display a critical understanding of a range of the principles, theories, concepts, terminologies and frameworks related to artificial intelligence and data engineering.	SC1	SCQF 9
LLO36	Demonstrate skills, attitudes, and behaviours appropriate to the workplace, including but not limited, to technical skills, communication skills, planning and organisational skills, and personal and professional skills.	SC1	SCQF 9
LLO37	Design and develop effective applications by using appropriate tools and techniques for a given requirement specification associated with artificial intelligence and data engineering.	SC2	SCQF 9
LLO38	Demonstrate applied knowledge of specialised skills in the development of intelligent systems using relevant deep learning techniques to solve real-world problems and awareness of strengths and limitations of such technology.	SC2	SCQF 9
LLO39	Design, implement and evaluate an Internet of Things (IoT) , Edge artificial intelligence robotics-based intelligent system using appropriate tools and techniques for a given requirement specification.	SC2	SCQF 9
LLO40	Apply theories, models, concepts and principles acquired during the course to the workplace.	SC2	SCQF 9
LLO41	Test and critically evaluate functionalities and security implications in the development of IoT, Edge AI or robotics-based autonomous system scenarios.	SC3	SCQF 9

LLO42	Demonstrate the ability to critically analyse, select and use principles and procedures systematically to real-world situations or problems related to deep learning and IoT , edge computing or robotics.	SC3	SCQF 9
LLO43	Critically appraise theory, process, solutions and outcomes based on experiences within the work-based learning environment.	SC3	SCQF 9
LLO44	Investigate professional problems and issues encountered when deploying IoT, Edge computing-based solutions or AI in robotics.	SC3	SCQF 9
LLO45	Recognise and reflect their own strengths and weaknesses as computing professionals in an organizational environment.	SC3	SCQF 9
LLO46	Demonstrate evidence of, and reflect on new learning of knowledge, skills and abilities acquired for effective practice in the IT industry.	SC4	SCQF 9
LLO47	Confidently and accurately select principles and procedures to solve a range of professional situations and problems, working with increasing autonomy in a work-based environment.	SC4	SCQF 9
LLO48	Demonstrate evidence of, and reflect on new learning with regard to knowledge, skills and abilities required for effective practice in the computing industry.	SC5	SCQF 9
LLO49	Maintain standards and quality control as applied to specification, production, implementation and maintenance of data engineering pipelines and AI-based systems/ data science applications, in accordance with current professional and ethical codes of practices.	SC5	SCQF 9
LLO50	Manage own learning, with minimal guidance, using a wide range of resources appropriate to the profession, seeking and making effective use of feedback.	SC5	SCQF 9
LLO51	Interact effectively within groups where appropriate, develop interaction and interpersonal communication skills with customers and other stakeholders.	SC5	SCQF 9
LLO52	Demonstrate an understanding of the context of, and relationship between, the principal areas, features, techniques, boundaries, terminology and conventions of artificial intelligence and data engineering.	SC1	SCQF 10
LLO53	Demonstrate confident familiarity and critical understanding of defining concepts and features of their chosen area of artificial intelligence and data engineering.	SC1	SCQF 10
LLO54	Demonstrate appropriate IT project management expertise, professional and ethical conduct, in-depth technical, problem solving and communication skills in the conduct of an individual research project.	SC1	SCQF 10
LLO55	Demonstrate a detailed knowledge and understanding of advanced specialisms in artificial intelligence and data engineering systems, including their theoretical foundations, applications, and real-world deployment contexts, awareness of innovation, commercialisation, and entrepreneurial applications of AI technologies.	SC1	SCQF 10
LLO56	Apply knowledge, skills and understanding in specialist areas of artificial intelligence and data engineering systems, including intelligent information retrieval, natural language processing, computational intelligence, machine vision, and their integration within real-world data-driven and deployed systems.	SC2	SCQF 10
LLO57	Apply practical knowledge of artificial intelligence and data engineering/ data science to execute a defined project of research, development, or investigation, demonstrating the ability to design, implement, and evaluate a complete end-to-end intelligent data-driven system.	SC2	SCQF 10
LLO58	Be able to keep up-to-date with the latest trends and applications in artificial intelligence and data engineering systems, demonstrating awareness of challenges arising from scalability, deployment, data quality, and real-world operational constraints.	SC2	SCQF 10
LLO59	Practice in a range of professional contexts that include a high degree of unpredictability and specialism in AI and data engineering systems, including system integration, deployment environments, and real-world data-driven applications.	SC2	SCQF 10
LLO60	Confidently and accurately select application of principles and procedures to solve a range of professional situations and problems, working autonomously, particularly within the individual research project.	SC3	SCQF 10

LLO61	Critically appraise and present current and emerging trends in artificial intelligence and data engineering systems, including their impact on scalable system design, deployment strategies, intelligent applications and innovation-driven or entrepreneurial opportunities in AI-based solutions.	SC3	SCQF 10
LLO62	Formulate and analyse complex problems in an increasingly independent, confident and flexible way, applying knowledge and skills appropriate to their solutions.	SC3	SCQF 10
LLO63	Critically review and consolidate knowledge, skills, practices and thinking in the domain of artificial intelligence and data engineering.	SC3	SCQF 10
LLO64	Demonstrate use and access a broad selection of specialist computing skills, tools and techniques for the individual research project.	SC4	SCQF 10
LLO65	Engage effectively in independent roles, produce professional critiques and reports, and communicate complex ideas, including those related to AI system development, innovation, and entrepreneurial applications of intelligent technologies.	SC4	SCQF 10
LLO66	Use effective means of communication to present, convey and express complex ideas and information, about specialised topics to inform audiences related to the individual research project, formally and informally.	SC4	SCQF 10
LLO67	Manage own learning with minimal guidance, using a wide range of resources while seeking and making effective use of feedback provided from academic supervisors particularly to the individual research project.	SC5	SCQF 10
LLO68	Maintain monitoring of standards and quality control as applied to the specification, production, implementation and maintenance of data engineering systems in accordance with current professional and ethical codes and practices.	SC5	SCQF 10

Programme Outcome Types Reference

#	SCQF Characteristic	SCQF Level
SC1	Knowledge and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC2	Practice: Applied knowledge, skills and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC3	Generic cognitive skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
		SCQF 7 - SCQF 7 SCQF 8 - SCQF 8

SC4	Communication, numeracy and IT skills	SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC5	Autonomy, accountability and working with others	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

Delivery Range

<i>Title</i>	<i>Description</i>
Full Time On Campus	

Stage 1 / Semester 1**Full Time On Campus**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM1606	Computational Mathematics [Approved]	Undertaken over two semesters.	Janani Harischandra	Yes	1.0	30	Level 7
Full Time On Campus	CM1604	Computer System Fundamentals [Approved]		Janani Harischandra	Yes	1.0	15	Level 7
Full Time On Campus	CM1603	Database Systems [Approved]		Janani Harischandra	Yes	1.0	15	Level 7
Full Time On Campus	CM1607	English Communication Skills [Approved]		Janani Harischandra	Yes	1.0	15	Level 7
Full Time On Campus	CM1601	Programming Fundamentals [Approved]	Undertaken over two semesters.	Janani Harischandra	Yes	1.0	30	Level 7

Stage 1 / Semester 2**Full Time On Campus**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM1602	Data Structures and Algorithms for Artificial Intelligence [Approved]		Janani Harischandra	Yes	1.0	15	Level 7

Full Time On Campus	CM1605	Web Technology [Approved]		Janani Harischandra	Yes	1.0	15	Level 7
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Stage 2 / Semester 1**Full Time On Campus**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE2607	Advanced Mathematics [Quality Event]		Janani Harischandra	Yes	1.0	15	Level 8
Full Time On Campus	CM2602	Artificial Intelligence [Approved]		Janani Harischandra	Yes	1.0	15	Level 8
Full Time On Campus	CE2603	Data Engineering Group Project [Post-Event Draft]	Undertaken over two semesters.	Janani Harischandra	Yes	1.0	30	Level 8
Full Time On Campus	CM2604	Machine Learning [Approved]		Janani Harischandra	Yes	1.0	15	Level 8

Stage 2 / Semester 2**Full Time On Campus**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM2606	Data Engineering [Approved]		Janani Harischandra	Yes	1.0	15	Level 8

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0757 - Electives 1	Artificial Intelligence and Data Engineering - Electives Group 1 [Academic Quality Scrutiny]		Janani Harischandra		1.0	15	
Full								

Time On Campus	0757 - Electives 2	Artificial Intelligence and Data Engineering - Electives Group 2 [Academic Quality Scrutiny]		Janani Harischandra		1.0	15	
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Stage 3 / Semester 1**Full Time On Campus**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM3604	Deep Learning [Approved]		Janani Harischandra	Yes	1.0	15	Level 9
Full Time On Campus	CM3606	Industrial Placement [Approved]	Undertaken over three semesters.	Janani Harischandra	Yes	1.0	90	Level 9

Stage 3 / Semester 2**Full Time On Campus**

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0757 - Electives 3	Artificial Intelligence and Data Engineering - Electives Group 3 [Academic Quality Scrutiny]		Janani Harischandra		1.0	15	

Stage 4 / Semester 1**Full Time On Campus**

Core								
Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
Full Time On Campus	CM4607	Computational Intelligence [Approved]		Janani Harischandra	Yes	1.0	15	Level 10
Full Time On Campus	CM4609	Individual Research Project [Post-Event Draft]	Undertaken over two semesters.	Janani Harischandra	Yes	3.0	60	Level 10
Full Time On Campus	CM4604	Research Trends [Approved]	Undertaken over two semesters.	Janani Harischandra	Yes	1.0	30	Level 10
Elective								
Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
Full Time On Campus	0757 - Electives 4	Artificial Intelligence and Data Engineering - Electives Group 4 [Academic Quality Scrutiny]		Janani Harischandra		1.0	15	

Course Deliveries

Stage	Semester	Delivery Type	Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
1	Semester 1	Core	Full Time On Campus	CM1606	Computational Mathematics [Approved]	Undertaken over two semesters.	Janani Harischandra	Yes	1.0	30	Level 7
1	Semester 1	Core	Full Time On Campus	CM1604	Computer System Fundamentals [Approved]		Janani Harischandra	Yes	1.0	15	Level 7
1	Semester 1	Core	Full Time On Campus	CM1603	Database Systems [Approved]		Janani Harischandra	Yes	1.0	15	Level 7

1	Semester 1	Core	Full Time On Campus	CM1607	English Communication Skills [Approved]		Janani Harischandra	Yes	1.0	15	Level 7
1	Semester 1	Core	Full Time On Campus	CM1601	Programming Fundamentals [Approved]	Undertaken over two semesters.	Janani Harischandra	Yes	1.0	30	Level 7
1	Semester 2	Core	Full Time On Campus	CM1602	Data Structures and Algorithms for Artificial Intelligence [Approved]		Janani Harischandra	Yes	1.0	15	Level 7
1	Semester 2	Core	Full Time On Campus	CM1605	Web Technology [Approved]		Janani Harischandra	Yes	1.0	15	Level 7
2	Semester 1	Core	Full Time On Campus	CE2607	Advanced Mathematics [Quality Event]		Janani Harischandra	Yes	1.0	15	Level 8
2	Semester 1	Core	Full Time On Campus	CM2602	Artificial Intelligence [Approved]		Janani Harischandra	Yes	1.0	15	Level 8
2	Semester 1	Core	Full Time On Campus	CE2603	Data Engineering Group Project [Post-Event Draft]	Undertaken over two semesters.	Janani Harischandra	Yes	1.0	30	Level 8
2	Semester 1	Core	Full Time On Campus	CM2604	Machine Learning [Approved]		Janani Harischandra	Yes	1.0	15	Level 8
2	Semester 2	Core	Full Time On Campus	CM2606	Data Engineering [Approved]		Janani Harischandra	Yes	1.0	15	Level 8
	Semester		Full Time		Artificial Intelligence and Data Engineering -		Janani				

2	2	Elective	On Campus	0757 - Electives 1	Electives Group 1 [Academic Quality Scrutiny]		Harischandra		1.0	15	
2	Semester 2	Elective	Full Time On Campus	0757 - Electives 2	Artificial Intelligence and Data Engineering - Electives Group 2 [Academic Quality Scrutiny]		Janani Harischandra		1.0	15	
3	Semester 1	Core	Full Time On Campus	CM3604	Deep Learning [Approved]		Janani Harischandra	Yes	1.0	15	Level 9
3	Semester 1	Core	Full Time On Campus	CM3606	Industrial Placement [Approved]	Undertaken over three semesters.	Janani Harischandra	Yes	1.0	90	Level 9
3	Semester 2	Elective	Full Time On Campus	0757 - Electives 3	Artificial Intelligence and Data Engineering - Electives Group 3 [Academic Quality Scrutiny]		Janani Harischandra		1.0	15	
4	Semester 1	Core	Full Time On Campus	CM4607	Computational Intelligence [Approved]		Janani Harischandra	Yes	1.0	15	Level 10
4	Semester 1	Core	Full Time On Campus	CM4609	Individual Research Project [Post-Event Draft]	Undertaken over two semesters.	Janani Harischandra	Yes	3.0	60	Level 10
4	Semester 1	Core	Full Time On Campus	CM4604	Research Trends [Approved]	Undertaken over two semesters.	Janani Harischandra	Yes	1.0	30	Level 10
4	Semester 1	Elective	Full Time On Campus	0757 - Electives 4	Artificial Intelligence and Data Engineering - Electives Group 4 [Academic Quality Scrutiny]		Janani Harischandra		1.0	15	

Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date
Sri Lanka	On-Campus	Full Time	

Sources of Further Information

No sources of further information recorded